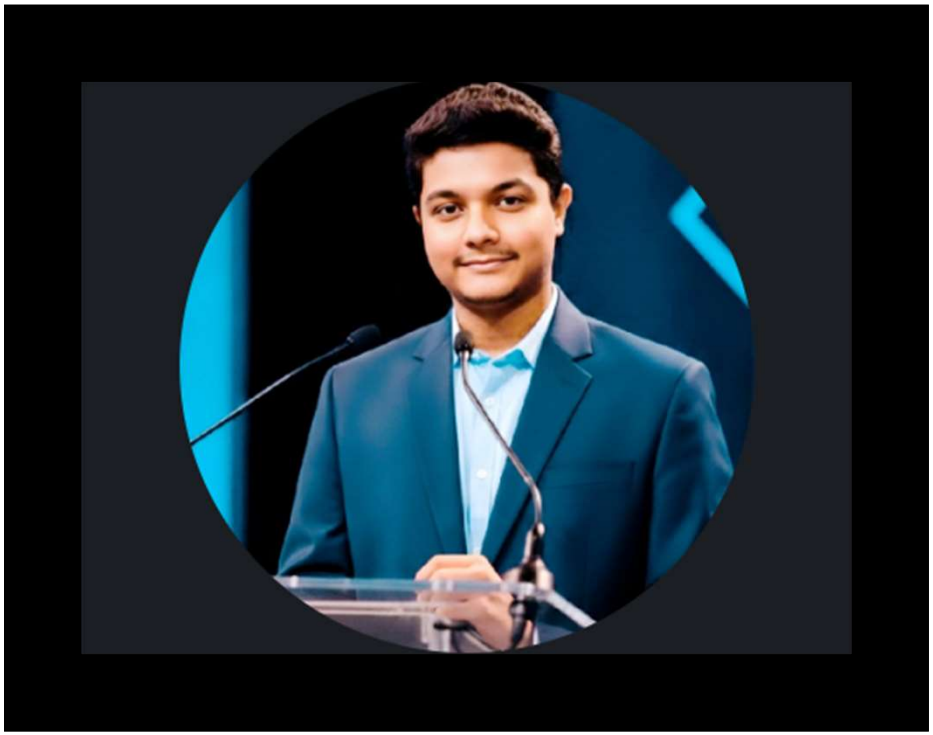
Microsoft Elevate

CAPSTONE PROJECT

Fintech Project Management



PRESENTED BY

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OUTLINE:

- **Problem Statement (Should not include solution)**
  Organizations struggle to manage multiple fintech projects across departments and regions. Inconsistent tracking of budgets, hours, and task completion leads to inefficiencies and delayed decisions.
- **Proposed System/Solution**
  A Power BI–based interactive dashboard that consolidates project, employee, and regional performance data to deliver actionable insights for fintech project management.
- **System Development Approach (Technology Used)**
  Power BI Desktop for report creation
  Power Query for data transformation
  DAX for calculated metrics
  Excel/CSV as data sources
  Power BI Service for deployment and sharing
- **Algorithm & Deployment**
  ETL process using Power Query → Data modeling with relationships → DAX measures for KPIs → Visualization layer with slicers and filters.
  Deployed via Power BI Service for real-time access and collaboration.
- **Result (Output Image)**
  Added in later slides.
- **Conclusion**
  The dashboard simplifies project tracking, highlights budget and completion trends, and empowers stakeholders to make faster, data-driven decisions across fintech operations.

PROBLEM STATEMENT:

- Managing many fintech projects across different departments and regions is not easy. Teams often face problems in tracking budgets, planned versus actual hours, and completion rates. This creates delays, wasted resources, and confusion.
- Without a clear and shared view of project progress, managers struggle to make quick decisions. Risks are not spotted early, and important opportunities are missed. Over time, this slows down delivery and reduces efficiency.
- A better system is needed — one that gives accurate, real-time insights into projects, helps predict demand for resources, and supports faster, smarter decision-making.

PROPOSED SOLUTION:

- The proposed system aims to address the challenge of predicting project demand patterns and resource requirements accurately by leveraging data analytics and machine learning. The solution will consist of the following components:
- **Data Collection**
  - Gather historical project and performance data, including budgets, timelines, completion rates, and resource allocation.
  - Enhance accuracy by incorporating real-time factors such as market conditions, risk levels, and departmental workload.
- **Data Preprocessing**
  - Clean and organize the collected data to handle missing values, outliers, and inconsistencies.
  - Apply feature engineering to highlight key factors that influence project success and resource demand.
- **Deployment**
- Develop a user-friendly Power BI dashboard that provides real-time insights into project status, budget utilization, and resource allocation.
- Deploy the solution on a scalable platform, ensuring fast response times and easy accessibility for managers and teams.

SYSTEM APPROACH:

- **System Requirements**
  - Historical project and performance data, including budgets, timelines, completion rates, and resource allocation.
  - Real-time inputs such as market conditions, risk levels, and departmental workload.
  - A platform capable of integrating multiple data sources and providing interactive visualizations.
- **Libraries / Tools Required to Build the Model**
- **Power BI Desktop** for dashboard creation.
- **Power Query** for ETL (extract, transform, load) and data preprocessing.
- **DAX** for advanced calculations and KPI measures.
- **Excel/CSV** as base data sources.
- **Python libraries** like Pandas, NumPy, and Scikit-learn for predictive modeling.

ALGORITHM & DEPLOYMENT:

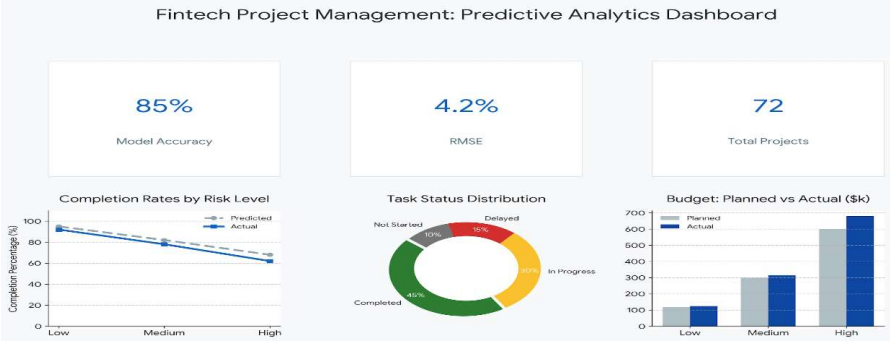
•**Algorithm Selection**
For predicting project completion rates and resource demand, a time-series forecasting model such as **ARIMA** or a deep learning model like **LSTM** is chosen. These algorithms are well-suited because they capture temporal patterns, seasonality, and risk-based variations in project performance, making them effective for forecasting timelines and budget utilization.

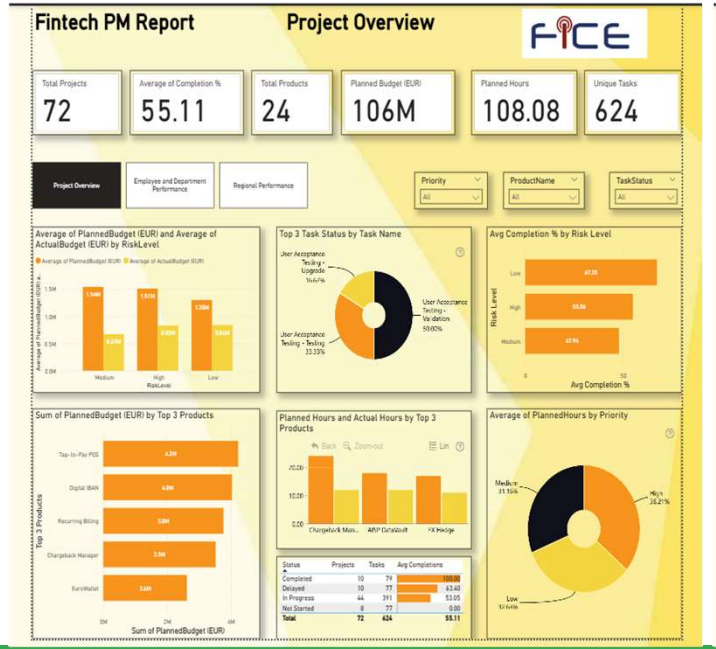
•**Data Input**
The model uses historical project data (budgets, timelines, completion rates, planned vs actual hours) along with external factors such as market conditions, risk levels, and departmental workload. These features help the algorithm understand both regular project progress and sudden deviations.

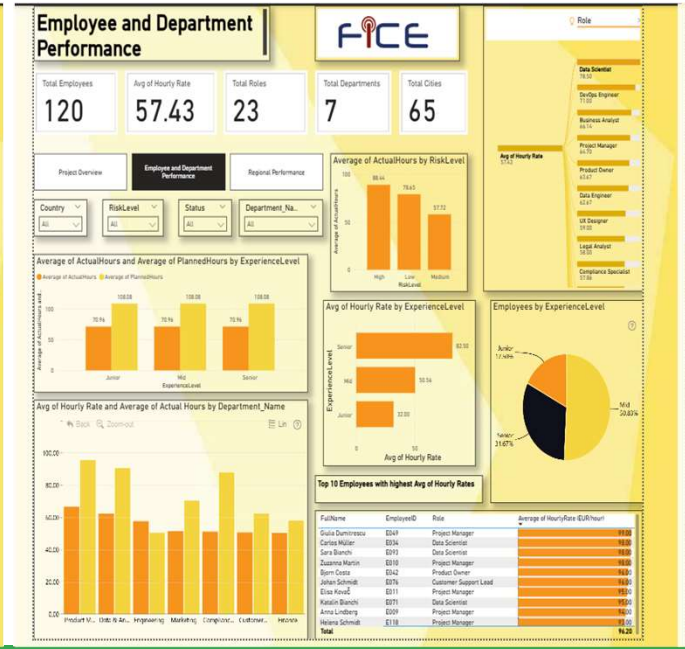
•**Training Process**
The algorithm is trained on historical datasets, applying techniques like **cross-validation** to ensure robustness. Hyperparameter tuning is performed to optimize accuracy, while preprocessing ensures clean and consistent input data. Feature engineering highlights critical drivers such as risk level, employee experience, and product type.

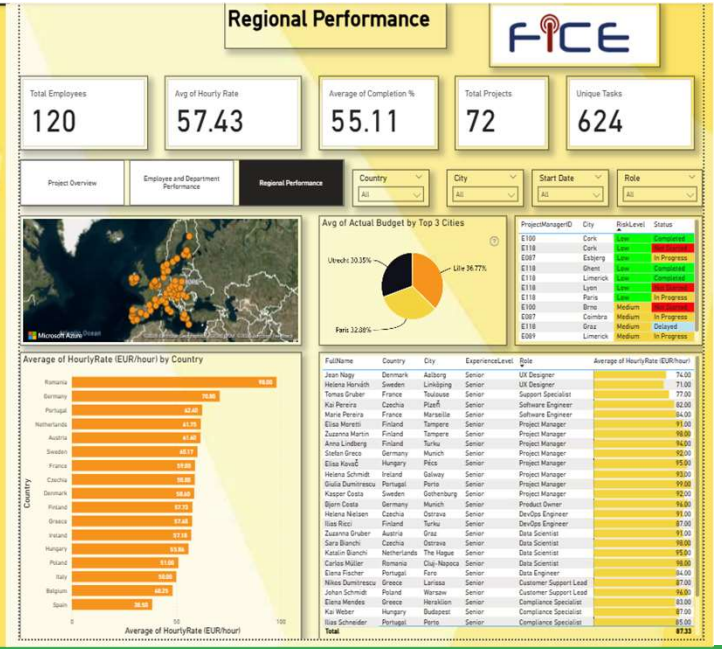
•**Prediction Process**
Once trained, the model generates forecasts for future project completion percentages, budget overruns, and resource demand. Real-time inputs such as updated market conditions or departmental workload are incorporated to refine predictions, ensuring the system adapts to changing business environments.

RESULT:


Fintech Project Management: Predictive Analytics Dashboard
85%
Model Accuracy
4.2%
RMSE
72
Total Projects
Completion Rates by Risk Level
Predicted
Actual
Completion Percentage (%)
Low
Medium
High
Task Status Distribution
Not Started
Delayed
In Progress
Completed
Budget: Planned vs Actual ($k)
Planned
Actual


Fintech PM Report
Project Overview
Total Projects
72
Average of Completion %
55.11
Total Products
24
Planned Budget (EUR)
106M
Planned Hours
108.08
Unique Tasks
624
Project Overview
Employee and Department Performance
Regional Performance
Priority
ProductName
TaskStatus
Average of PlannedBudget (EUR) and Average of ActualBudget (EUR) by RiskLevel
Top 3 Task Status by Task Name
Avg Completion % by Risk Level
Sum of PlannedBudget (EUR) by Top 3 Products
Planned Hours and Actual Hours by Top 3 Products
Average of PlannedHours by Priority


Employee and Department Performance
Total Employees
120
Avg of Hourly Rate
57.43
Total Roles
23
Total Departments
7
Total Cities
65
Average of ActualHours by RiskLevel
Average of ActualHours and Average of PlannedHours by ExperienceLevel
Avg of Hourly Rate and Average of Actual Hours by Department_Name
Avg of Hourly Rate by ExperienceLevel
Employees by ExperienceLevel
Top 10 Employees with highest Avg of Hourly Rates


Regional Performance
Total Employees
120
Avg of Hourly Rate
57.43
Average of Completion %
55.11
Total Projects
72
Unique Tasks
624
Avg of Actual Budget by Top 3 Cities
Average of HourlyRate (EUR/hour) by Country

CONCLUSION:

- The proposed system effectively demonstrates how data analytics and machine learning can enhance project management in the fintech domain. By integrating historical project data with real-time factors such as market conditions, risk levels, and resource availability, the solution provides accurate forecasts of completion rates, budget utilization, and resource demand.
- The Power BI dashboard deployment ensures that these insights are accessible in real time, enabling managers to make informed decisions and respond quickly to risks. Challenges encountered during implementation included handling inconsistent data, tuning models for diverse project types, and ensuring scalability for enterprise use.
- Future improvements could involve incorporating more external market signals, experimenting with advanced deep learning models for better accuracy, and refining the dashboard interface for faster decision support.
- Accurate predictions are crucial in fintech project management, as they help maintain financial stability, optimize resource allocation, and ensure timely delivery of products in a highly competitive industry.

FUTURE SCOPE:

- **Potential Enhancements & Expansions**
- **Additional Data Sources**
  - **Integrate external market signals such as financial indices, regulatory changes, and competitor activity.**
  - **Include HR and resource data (employee availability, skill levels, hourly rates) to improve accuracy in predicting project delays and budget overruns.**
- **Algorithm Optimization**
  - **Experiment with advanced machine learning models (e.g., LSTM, Gradient Boosting, or ensemble methods) to capture complex project dynamics.**
  - **Apply hyperparameter tuning and automated model selection to improve forecasting accuracy.**
- **Geographic Expansion**
  - **Extend the system to cover multiple cities or regions, allowing cross-regional comparisons of project performance.**
  - **Incorporate regional economic indicators and local risk factors for more context-aware predictions.**
- **Emerging Technologies**
- **Edge Computing: Enable real-time processing of project data closer to source systems, reducing latency and improving responsiveness.**

•**Advanced ML Techniques: Explore reinforcement learning for adaptive resource allocation or anomaly detection to flag unusual** project behaviors.
•**Cloud Integration:** Scale dashboards and predictive models seamlessly across departments and geographies.

REFERENCES:

List and cite relevant sources, research papers, and articles that were instrumental in developing the proposed solution. This could include academic papers on bike demand prediction, machine learning algorithms, and best practices in data preprocessing and model evaluation.

GitHub Link: https://github.com/Gyanankur23/Fintech-PM-Forecasting

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Thank You